

UMF — Formal Framework

Unified Metric Framework, current structural commitments and formal scaffold

Stephen L Browne | Central Coast NSW, May 2026 (AEST) | v2.3 | Supersedes v2.2 (18 May 2026)

Preamble

This document records the current commitments of the Unified Metric Framework (UMF). It is intended for specialist review and is the document on which the specialist engagement is expected to focus. It supersedes UMF v2.2 of 18 May 2026, with the substantive changes documented in the revision history at the end.

The framework is now publicly available at <https://github.com/OldGreyWhistle/The-UMF> as a citable reference. The current package is V2.3. Previous published versions (V2.0, V2.1, V2.2) are preserved in the repository for revision history.

Version 2.3 incorporates housekeeping refinements consequent on public GitHub publication and small additions consequent on the post-V2.2 commentary. The substantive additions: an open theoretical question on entropy behaviour at the reflective wavefront (raised by the ChatGPT comment on V2.2); a consolidated open theoretical question on DM scaling with cosmic time and its potential population-level early-universe consequences, coupled to the variable- α / BBN engagement and the S_8 / CMB acoustic peaks unification question; explicit citation of the framework lineage's ISW meta-analysis constraint ($|\alpha| < 0.3$ on linear redshift trend in A_{ISW}); reference to the bounded-universe topology positioning document v1.0; the GitHub repository reference throughout; and location and timezone consistency (Central Coast NSW, AEST in standard-time months). No structural commitments are changed in v2.3.

The framework is a pre-paradigmatic structural picture, not a developed theory. It makes no quantitative predictions that distinguish it from Λ CDM at the framework's rigour standard (5σ for proofs, 4σ for correlations). At lower confidence (2σ - 3σ notable), the framework can identify directions of prediction that are not in tension with current data, but these are not claims of empirical support. Near-term observational programmes are listed as the contexts in which future quantitative content would have to make contact with data.

The framework should be read in conjunction with the Foundational Logic document, which states the underlying structural commitments, and the Empirical Engagement document, which presents the observational programme. The covering note provides provenance and the specific deliverable being requested from the specialist. The Bounded-Universe Topology positioning document (v1.0) is a companion piece engaging the framework's reflective-boundary commitment with the Planck CMB low- ℓ literature at the structural-inference level.

Standing rules

The framework operates under three standing rules that govern how commitments are made and evaluated. These rules have been developed across the framework's revisions and are integral to its current state.

Rigour standard

The empirical bar for findings is 5σ for proofs and 4σ for correlations. Findings at 2σ are notable only, meaning they are worth recording but they are not evidence. Findings at 3σ are in the grey zone — interesting, worth investigation, not a result. This is the standard particle-physics and cosmology bar for these categories and is the correct bar for work that is asking to be taken seriously. The framework applies this standard to its own claims and to its citations of external data: findings at 2σ - 3σ are noted as consistency or directional alignment without claiming support; findings at 4σ or above can be cited as evidence.

Honesty in evaluation

The framework is to be evaluated by what it commits to and what it does not commit to, presented honestly rather than for rhetorical effect. Negative results are reported alongside positive findings, with both presented in plain language. Speculative content is flagged as speculative; settled content is flagged as settled. Candidate commitments are flagged as candidates pending evaluation, not as adopted positions.

Full evaluation in UMF context before import

Any formalism, metric, or theoretical apparatus imported into UMF from existing physics must be evaluated in full before adoption. The evaluation must include the units of every quantity involved, the regime of validity of the imported formalism, the assumptions underlying it in its original context, and the specific way it maps onto UMF's structural commitments. No formalism is imported on the basis of structural analogy alone. Candidate imports are flagged as candidates pending evaluation, not as adopted apparatus.

Permanent commitment on c

The framework commits to the invariance of the speed of light c . Any mechanism that requires c -variation is, by that requirement, incompatible with the framework as currently constituted. Variable c breaks spacetime structure by definition and any future synthesis-pull toward variable c is to be caught by this rule. This commitment is recorded permanently and is not subject to revision within the framework's structural commitments.

Structural commitments

The bounded universe and its regions

The framework treats the universe as a bounded structure with three identifiable regions. Region A is the established spacetime within the wavefront — the universe we observe, with its matter content, its metric structure, and its dynamical evolution. Region B is the wavefront itself, a finite-thickness buffer through which gravitational tension transitions from its inner value to its outer value. Region C is the primordial state outside the wavefront, characterised by gravitational equilibrium and not admitting the conventional categories of spacetime, energy, or matter.

The wavefront is the active boundary that separates Region A from Region C. It is not a passive interface but a structural feature with its own dynamics. The buffer transitions T_G (gravitational tension) from its inner value to the equilibrium value of zero in Region C. The inner surface of the wavefront carries the cosmological surface tension at the maximum sustainable level, with the magnitude given by the Planck-scale dimensional construction $P_P \times I_P$.

The wavefront as totally reflective inner boundary

The wavefront's inner surface is totally reflective to all properties of the established spacetime regime — electromagnetic radiation, gravitational waves, matter, particles. The only quantity that crosses the boundary is gravitational tension T_G itself, in the direction of relaxation toward primordial equilibrium. This commitment is operational for the framework's wholly-gravitational relationship between Region A and Region C: spacetime content is bounded inside Region A, and the boundary admits only the gravitational displacement that established the regime.

The reflectivity is total (perfect mirror) and constant across the established phase of the cosmic lifecycle. The reflectivity terminates only when the wavefront's structural integrity fails at the terminal phase. The framework does not commit to partial or evolving reflectivity; if the empirical engagement produces tension with the totally-reflective commitment, the question of partial reflectivity can be reopened, but the default position is total reflection.

This commitment is inherited from the framework lineage (SS-TLGO design document, April 2026; Prime-Ordeal blog document, January 2026; UMDT and UMG T precursors). It is the operational form of Postulate 3 of the foundational logic. The commitment was not explicitly carried into v2.1 of this document; v2.2 restored it and v2.3 retains it.

The reflective-boundary commitment has consequences for the framework's observational predictions. CMB photons reflect at the wavefront, producing potential signatures at low ℓ if Region A is small enough relative to the surface of last scattering for round-trip paths to be observable. Primordial gravitational waves from the establishing event reflect inside Region A, producing a stochastic gravitational-wave background internal to Region A (see below). The empirical engagement document and the bounded-universe topology positioning document develop these observational consequences.

The gravitational tension field T_G

The framework commits to T_G as a dynamical field with units of pressure (force per area, same units as the Planck pressure). T_G characterises the gravitational stress at each point in Region A. T_G is bounded above by the Planck pressure P_P within Region A under sustained conditions; it can exceed P_P during the initial establishing event (the trans-Planckian initial regime) when the displacement from primordial equilibrium is greatest and before baryonic matter creation has damped the initial pulse.

T_G is the framework's load-bearing dynamical variable. Its variation across Region A drives the framework's cosmological dynamics; its variation over the cosmic lifecycle characterises the framework's lifecycle structure (trans-Planckian initial, sustainable plateau, relaxation, terminal). The environmental couplings of the electromagnetic vacuum (ϵ_0 and μ_0) vary with local T_G value, producing observable signatures.

The framework adopts the option-B interpretation of T_G from the V1.5 cold review: T_G is an independent dynamical field with its own field equation. This reverses the V1.5 commitment to option A (T_G as algebraic scalar derived from $T_{\mu\nu}$). The reason for the reversal is that the option-A interpretation does not support the framework's commitment to T_G as the variable driving variable-coupling dynamics. The option-B interpretation requires the framework to commit to a Lagrangian and a coupling prescription for T_G , which is theoretical work that the framework cannot yet specify and that is the gating issue for further development (see Open theoretical questions below).

The Planck-scale physical maxima

The framework treats the Planck force F_P and the Planck pressure P_P as physical maxima within Region A under sustained conditions. F_P is the maximum allowable force; P_P is the maximum allowable internal stress and the maximum sustainable T_G in the sustainable regime. The cosmological surface tension at the wavefront's inner surface is given by the dimensional construction $P_P \times l_P$ with units of N/m (force per length, the dimensional form of surface tension). The dimensional identity $P_P \times l_P^2 = F_P$ holds exactly and is verified.

The Planck-scale anchors have been numerically verified against CODATA-2018 constants. The framework's values are within 0.2% of the recomputed values, well inside the propagated CODATA uncertainty on G (~22 ppm). The numerical anchors are not at issue. What is at issue is whether these quantities have the physical roles the framework assigns them. The framework's commitment is that Planck-scale quantities function as physical maxima rather than as scales where physics changes character; this commitment is structural and has been flagged in cold review as requiring justification from deeper theory.

Quantity	Value (units)	Physical role in UMF
Planck force $F_P = c^4/G$	1.21×10^{44} N (force)	Maximum allowable force within Region A under sustained conditions
Planck pressure $P_P = c^7/(\hbar G^2)$	4.63×10^{113} N/m ² (pressure)	Maximum sustainable internal stress within Region A; bound on T_G under sustained conditions
Cosmological surface tension $P_P \times l_P$	$\approx 7.5 \times 10^{78}$ N/m (surface tension)	Maximum sustainable surface tension at wavefront inner surface; load-bearing for wavefront structural integrity and reflective property

Under the initial establishing event, T_G exceeds P_P . This is the trans-Planckian regime, where the framework explicitly does not commit to specific physics because standard formalism breaks down at these conditions. The trans-Planckian regime is invoked only where it does specific work — at the initial establishing event itself — and is not used as a generic escape valve elsewhere in the framework. Where the framework does not specify physics, the documentation states "not specified by the framework" rather than invoking trans-Planckian regimes.

Constant gravitational coupling G and constant c

The framework commits to G being constant at its observed value (the standard CODATA value, with the value being a constant of the established spacetime regime rather than a variable of state). This commitment supersedes the V1.5 commitment to variable G , which was a compromise to accommodate dark matter phenomenology and which produced an unresolved Bianchi-identity consistency issue. The current dark matter formalism (see below) does not require variable G , and constant G defuses the Bianchi-identity issue.

The framework commits to c being constant at its standard value. This is permanent (see standing rules above) and is not subject to revision. The Einstein equations apply in standard form within Region A, with the modifications introduced by the framework being in the electromagnetic vacuum coupling sector (ϵ_0 and μ_0) and in the gravitational stress field T_G , not in the gravitational coupling G or the propagation speed c .

Variable electromagnetic vacuum couplings ϵ_0 and μ_0

The framework commits to the electromagnetic vacuum properties varying as environmental variables of the gravitational state. Specifically, ϵ_0 varies with the local value of T_G , and μ_0 varies inversely to ϵ_0 to preserve the c-invariance under the exact identity $\epsilon_0 \mu_0 c^2 = 1$. The c-invariance commitment forces this inverse variation; there is no independent choice about whether μ_0 varies.

The fine structure constant α depends on ϵ_0 (and equivalently on μ_0 in the equivalent formulation), so variable ϵ_0 produces variable α . This is the connection point to existing empirical programmes that have searched for α -variation — quasar absorption spectroscopy and atomic clock comparisons. The framework predicts variable α as a direct consequence of variable ϵ_0 , with the magnitude of variation determined by the specific functional form of $\epsilon_0(T_G)$.

The specific functional form of ϵ_0 as a function of T_G is not specified by the framework. This is theoretical work that requires either derivation from the framework's deeper commitments or empirical fitting to observed signatures. The framework flags this as an open question requiring specialist input. Both cold reviews (ChatGPT and Claude Opus 4.7 incognito) recommend that the variable-coupling sector be the first quantitative work item the framework attempts, using the Bekenstein 1982 / BSBM (Sandvik–Barrow–Magueijo 2002) / Olive–Pospelov 2002 formalism as candidate apparatus subject to evaluation under the framework's import rule.

The dissolving polar shell as candidate relaxation picture

The framework adopts, as a candidate structural picture for the wavefront's dynamics, the dissolving polar shell analogy. The wavefront is treated as a shell with structural integrity maintained by surface tension at the cosmological surface tension scale ($P_P \times I_P$) on its inner face. The shell relaxes over the cosmic lifecycle, with the relaxation being the process that produces the observed dark-energy phenomenology. The polar character of the shell (in the analogy) corresponds to the asymmetric properties of the buffer across its thickness — different conditions at the inner and outer surfaces.

The plateau phase of the lifecycle corresponds to the shell's surface tension being sustained at the cosmological surface tension scale with slow relaxation. The relaxation phase corresponds to the surface tension dropping below the threshold required to sustain the maximum surface tension level, with the relaxation rate increasing as the shell's structural integrity decreases. The terminal phase corresponds to the shell having relaxed sufficiently that the bounded spacetime cannot be sustained.

The formal apparatus for the dissolving shell picture is an open question. The candidate physical pictures connect to existing quantum gravity programmes (loop quantum gravity, causal set theory, holographic approaches, asymptotic safety, post-Hawking black hole information work), each of which provides different formalism for boundary dynamics at near-Planck-scale stress. The Israel junction conditions (Israel 1966) and brane cosmology formalism (Randall–Sundrum 1999 and successors) provide candidate formalism for the wavefront as a structural object, with contemporary peer-reviewed applications (Schubert 2026 in Entropy uses Israel junction conditions on configurations structurally similar to UMF). The framework cannot specify which programme's apparatus is most aligned with its commitments without specialist input. The dissolving shell picture should be read as a candidate structural model whose formal apparatus is open, not as a committed formalism.

Dark matter as gravitoelectric field contribution

The framework commits to single-species dark matter characterised as a no-mass, no-particles field effect, observable through deflection signatures on test particles and through gravitational lensing of electromagnetic radiation. Dark matter is a property of the gravitational medium, sourced by the wavefront and modulated locally by the gravitational distribution of baryonic matter (galactic concentrations, including supermassive black holes where present). Under the reflective-boundary commitment, the source mechanism is not particles arriving from outside — particles cannot cross the boundary — but the framework's gravitational coupling between the wavefront's structural state and the gravitoelectric field internal to Region A.

The formal characterisation of dark matter is through its contribution to the gravitoelectric field — the analogue of the electric field in the gravitoelectromagnetism (GEM) correspondence. The gravitoelectric field is the standard gravitational acceleration field characterising how test particle trajectories deviate at each point. The framework's dark matter contributes to this field without being a mass-energy source for it. The standard mass-energy sources (baryonic matter through $T_{\mu\nu}$) continue to source the gravitoelectric field through the standard Einstein equations; the framework's dark matter adds a separate contribution that follows from T_G dynamics and the wavefront sourcing.

The dark matter contribution to the gravitoelectric field has two distinct components. The static component is the field configuration in regions where the gravitational driving conditions are not time-varying; this component produces the rotation curve and lensing signatures already observed and attributed to dark matter in standard cosmology. The time-derivative component (analogous to displacement current in electromagnetic theory) appears in regions where the gravitational driving conditions are themselves time-varying — particularly in supermassive black hole environments with accretion dynamics, jet launching, and time-varying orbital motion. The time-derivative component produces electromagnetic-propagation signatures distinct from the gravitational deflection signature.

The frame-dragging (gravitomagnetic) component of GEM does not apply to the framework's dark matter, because dark matter has no mass-energy current to source it. Frame-dragging continues to apply to baryonic matter under standard physics.

The specific formalism for the gravitoelectric dark matter contribution is at the level of candidate. The candidate formalisms include scalar-tensor theories (Brans-Dicke and descendants, Horndeski with cubic and shift-symmetric sectors viable post-GW170817), TeVeS (Bekenstein 2004), MOG / Scalar-Tensor-Vector Gravity (Moffat 2006 onward), and Verlinde's emergent gravity (2016). The framework cannot commit to a specific candidate without specialist evaluation. The structural commitment is to gravitoelectric field characterisation; the specific mathematical formulation is open.

Black holes as local wavefront instances

The framework treats black holes as local instances of the same boundary structure that the global wavefront exhibits. The framework's commitment is structural: a black hole horizon is a boundary structure with the same general properties as the cosmic wavefront, scaled to the black hole's specific size and mass. The relaxation dynamics of the cosmic wavefront should govern the black hole horizon's dynamics with appropriate scaling. The Claude Opus 4.7 cold review correctly flagged that "Planck-pressure boundary" cannot mean literally Planck pressure for an astrophysical SMBH, where the local stress at the horizon is many orders of magnitude below Planck pressure; the commitment is that the structural character is the same, with the

specific values being scale-dependent. The specific scaling relationship is open theoretical work.

This commitment provides a tractable intermediate case for testing the framework's wavefront dynamics. Black holes are observationally accessible, their properties (mass, spin, area) are measurable, and substantial theoretical work has been done on their boundary dynamics (the Hawking radiation calculation and its extensions, the black hole information literature). A specialist with quantum gravity background could in principle work out the framework's prediction for black hole horizon dynamics and check it against existing observations and existing theory, before attempting the more difficult cosmic wavefront calculation.

Cosmic lifecycle

The framework commits to a finite cosmic lifecycle structured by the dynamics of T_G and the wavefront. The phases of the lifecycle are characterised by T_G 's value and the wavefront's structural state. The establishing event is named "blast metrics" as an analogical framing for the high-stress regime where wave and energy components of the primordial pulse are bound together. The framework's working commitment is that wave and energy components propagate together during this regime, with the deeper physics of wave-energy distinction left as an open theoretical question.

Phase	T_G state	Wavefront state	Observable signature
Establishing event (blast metrics)	Trans-Planckian, damped by BAO creation	Forming, surface tension establishing	Not directly observable; corresponds to standard "Big Bang"
Plateau	Sustainable, bounded by P_P	Surface tension at maximum sustainable level, slow relaxation; reflective inner boundary established	Standard cosmological observations; Λ CDM-compatible
Relaxation (candidate current epoch)	Beginning to decline below maximum sustainable	Surface tension dropping below maximum sustainable level; reflectivity maintained	Dark energy phenomenology; monotonic $w(z)$ deviation
Terminal	Returning to zero (primordial equilibrium)	Dissolved; reflectivity terminates	Not directly observable; end of cosmic lifecycle

The framework does not currently commit to a specific identification of the current epoch within the lifecycle. The current epoch is plateau or early relaxation; the framework cannot distinguish between these at present. If the current epoch is plateau, the observed acceleration of cosmic expansion is at the resolution limit of current observations and is consistent with cosmological-constant behaviour ($w = -1$) to the precision DR2 can measure. If the current epoch is early relaxation, the observed acceleration is the early signature of the relaxation process beginning, with $w(z)$ just starting to deviate from -1 .

DESI DR3, expected in 2026-2027, is the decisive empirical test. If the DR2 signal strengthens to 4σ correlation-class evidence in the direction the framework predicts (see Empirical Engagement document), the framework can commit to early relaxation as the current epoch. If the signal remains at notable level, plateau remains viable. If the signal disappears or reverses, the framework's position requires revisiting.

The framework predicts monotonic $w(z)$ in the relaxation phase — the equation-of-state parameter does not oscillate or reverse during the observed redshift range. This is the framework's primary near-term falsifiable prediction. The specific functional form of $w(z)$ during the relaxation phase is not specified by the framework; the prediction is that the dependence is monotonic, with the specific shape being open.

Primordial gravitational signature under the reflective-boundary commitment

The framework's commitment to a totally reflective inner boundary on the wavefront has specific consequences for the gravitational signature of the establishing event. Primordial gravitational waves from the establishing event do not propagate freely out to infinity (which is impossible in the framework's bounded universe regardless of the boundary character); they reflect off the wavefront's inner surface and remain bounded inside Region A.

The framework's commitment is that the primordial gravitational content inside Region A is stochastic rather than coherent. The establishing event is a single chaotic event producing a stochastic content distribution, not a coherent periodic source. The reflected gravitational content therefore forms a stochastic gravitational-wave background internal to Region A rather than a coherent standing-wave structure. This is the framework's position on what was previously framed as the "monopolar gravitational-wave echo" in earlier framework versions.

The stochastic-background framing partially defuses the Birkhoff theorem issue that the Claude Opus 4.7 cold review flagged. Standard GR's Birkhoff theorem forbids spherically symmetric gravitational radiation from a localised source in vacuum. A stochastic gravitational-wave background is not spherically symmetric in the relevant sense — individual modes have arbitrary angular structure, and the stochastic character means the background is statistically isotropic rather than dynamically spherically symmetric. The framework's commitment to stochastic content rather than coherent monopolar content sidesteps the Birkhoff prohibition cleanly.

The reformulation in scalar-tensor language may still be needed for the establishing pulse itself (which is not a vacuum solution) and for the wave-energy distinction in the blast-metrics regime, but the stochastic-background framing is consistent with standard GR for the post-establishment regime. The framework's observational predictions for gravitational-wave observations should be at the stochastic background level rather than at the coherent-mode level.

Open theoretical questions

The framework has several open theoretical questions that are flagged for specialist input. These are not flaws; they are the work items where the specialist's expertise is most needed and where the framework's development cannot proceed without specialist contribution. The first question is the gating issue: without an action principle for T_G , the framework cannot produce quantitative predictions, cannot

check conservation laws, and cannot evaluate stability. Both cold reviews identified this as the load-bearing theoretical priority.

- **Gating issue: action principle for T_G .** The framework commits to T_G as an independent dynamical field (option B from the V1.5 cold review). Specifying the action principle, Lagrangian, and coupling prescription for T_G is the gating issue — without this, the framework cannot produce quantitative predictions, cannot check conservation laws, and cannot evaluate stability. Both cold reviews recommend addressing this through scalar-tensor formalism (Horndeski with the post-GW170817 viable sector, beyond-Horndeski as needed) with T_G identified either with a Horndeski scalar ϕ or with a derived quantity. The specialist input should address what action T_G takes, how it couples to standard physics, and whether it produces predictions consistent with cosmological observations.
- **Variable-coupling Maxwell equations and $\epsilon_0(T_G)$ functional form.** Both cold reviews recommend this as the first quantitative work item, using the Bekenstein 1982 / BSBM (Sandvik–Barrow–Magueijo 2002) / Olive–Pospelov 2002 formalism. The work can proceed without committing to a full gravity theory: T_G is treated as an unspecified scalar field, the Bekenstein action couples it to the electromagnetic kinetic term, predicted α evolution is calculated, and the result is compared against atomic-clock bounds ($d \ln \alpha / dt < 10^{-17}/\text{yr}$), quasar spectroscopy (ESPRESSO sub-ppm bound, Murphy 2022), and BBN constraints. The framework's variable- ϵ_0 commitment either survives this comparison or is falsified. This is the most tractable empirical engagement the framework can attempt.
- **The relaxation mechanism producing $w \approx -1$ behaviour.** Both cold reviews identify this as the framework's most serious physical gap. Boundary relaxation of content from a bounded region normally produces dilution of energy density inside the region, not negative effective pressure. The observed dark-energy phenomenology is $w \approx -1$, which behaves as if energy density does not dilute with expansion. The framework needs to specify the mechanism by which wavefront relaxation produces an effective equation of state with w close to -1 rather than producing accelerated dilution. The reflective-boundary commitment is relevant here: the relaxation that crosses the boundary is gravitational tension itself, not energy in the conventional sense, so the standard dilution argument may not apply in the same form. This connection requires specialist examination.
- **Formal apparatus for dark matter as gravitoelectric field contribution.** The framework commits to the gravitoelectric field characterisation but does not specify the formal apparatus. Candidate formalisms include scalar-tensor theories, MOND in relativistic extensions (TeVeS, MOG), and Verlinde's emergent gravity. The specialist input should address which of these is most aligned with the framework's commitments. Claude Opus 4.7 cold review intuited (offered as intuition, not as evaluation) that MOG is the most actively developed at present and may be the best place to start a comparative read.
- **DM scaling with cosmic time and population-level early-universe consequences.** UMF's gravitoelectric DM has unspecified scaling with cosmic time. Standard cold DM scales as a^{-3} by mass conservation; UMF's wavefront-sourced gravitoelectric contribution may have the same or different scaling. The scaling matters for the framework's predictions at recombination, BBN, and later epochs, and should be specified by the specialist work as part of the gravitoelectric DM formal apparatus. A potential consequence of higher early-universe DM density, if UMF reproduces standard scaling, is population-level effects on early stellar evolution including marginally shifted initial mass functions and supernova timing as a function of cosmic time. Whether these consequences are distinguishable from standard astrophysical uncertainties requires the specialist work to produce specific quantitative predictions. This question is coupled to the variable- α / BBN engagement and to the S_8 clumpiness / CMB acoustic peaks unification question below; the three should be evaluated together.

- **Potential simultaneous resolution of S_8 clumpiness deficit, BBN light-element abundances, and CMB acoustic peaks structure.** The framework's commitment to gravitoelectric DM with wavefront sourcing modulated by baryonic concentrations may produce DM behaviour at recombination that differs from cold collisionless DM behaviour, with cascading effects on inferred cosmological parameters and BBN abundance predictions. The S_8 tension at $2-3\sigma$ in weak lensing, the BBN light-element abundance anomalies (lithium-7 deficit, possible primordial helium issues), and the CMB acoustic peaks dependence on DM clustering at recombination are observationally distinct but may share a common dependence on DM behaviour. The specialist input should address whether UMF's commitments produce a coherent simultaneous resolution of these tensions, or whether the same trade-offs that affect other modified-DM proposals (MOND, TeVeS, MOG, fuzzy DM, warm DM) also affect UMF. This question is recorded as a potential area for investigation rather than as a framework commitment.
- **DM behaviour at recombination (CMB acoustic peaks).** The framework's gravitoelectric DM "sourced by the wavefront and modulated locally by baryonic concentrations" presupposes structures that do not exist at $z \approx 1100$. Galaxies and SMBHs are not yet formed at recombination. The CMB acoustic peak structure (Planck 2018, ACT, SPT) requires non-baryonic dark matter to be gravitationally clustering by recombination. The framework needs to specify how the gravitoelectric DM contribution behaves at recombination. This is the standard hard test that modified-gravity / no-particle DM programmes face. The author has indicated existing UMDT-era CMB analysis work that may be relevant to this question; this is to be incorporated in a future revision once summarised.
- **Stochastic gravitational-wave background under reflective boundary.** The framework's commitment to a stochastic gravitational-wave background internal to Region A (sourced by the primordial pulse and reflected within Region A by the wavefront) requires formal apparatus that the framework does not yet specify. Specifically: the spectrum of the SGWB sourced by the establishing pulse, the modification of the spectrum by reflection at the wavefront, and the relationship to potentially observable signatures. The Birkhoff theorem issue is partially defused by the stochastic framing, but the scalar-tensor or beyond-GR reformulation may still be needed for the establishing pulse itself and for the wave-energy distinction in the blast-metrics regime.
- **Entropy behaviour at the reflective wavefront.** The framework's commitment to total reflection at the wavefront's inner surface produces specific entropy questions inherited from the black hole information theory literature. The specialist input should address what entropy accounting is consistent with the reflective-boundary commitment (does information get scrambled at the boundary, does the reflection preserve entropy, what is the von Neumann entropy of the bounded region under the reflective-boundary commitment), and how the framework's gravitational-budget conservation relates to entropy conservation. This question was raised in the post-V2.2 commentary (ChatGPT, May 2026) and is recorded here as an open theoretical question for specialist evaluation; the framework does not currently take a position on the answer.
- **Wavefront sourcing mechanism for T_G .** The framework commits to the wavefront sourcing the gravitoelectric DM contribution through T_G dynamics but does not specify the sourcing mechanism. Candidate mechanisms include the wavefront's geometric structure, its surface tension, its relaxation rate, or some combination. Each would produce different empirical predictions. The specialist input should address what mechanism is most consistent with the framework's structural commitments and what observable signatures would distinguish the candidates.
- **Variable-coupling Maxwell equations consistency.** Maxwell's equations with variable ϵ_0 and μ_0 produce source-like terms from the gradients of the variable constants. The framework needs a specific formulation that is internally consistent. The Bekenstein/BSBM apparatus provides one such formulation; the specialist input should

address whether this formulation is the right one for UMF and what its consequences are for electromagnetic phenomenology in regions of variable T_G .

- **Primordial equilibrium reference state and Postulate 1.** The framework commits to Region C as the primordial gravitational equilibrium, characterised by zero gravitational tension and not admitting conventional categories of spacetime, energy, or matter. Claude Opus 4.7 cold review flagged "gravitational equilibrium without spacetime" as a category error in standard GR terms, requiring commitment to a specific pre-geometric programme (LQG, causal set theory, holographic approaches) to do real work. The specialist input should address either (a) which pre-geometric programme provides the ontology of the primordial state, or (b) whether Postulate 1 can be restricted to a definitional role with no subsequent commitment depending on its substance.
- **Static-to-dynamic dark matter transition.** The framework commits to the time-derivative component of the dark matter contribution being present in regions of time-varying gravitational driving (SMBH environments). The specific threshold or functional dependence of the static-to-dynamic transition is not specified. The specialist input should address what conditions are sufficient to drive the dark matter into the dynamic regime and what observable signatures distinguish the dynamic component from the static component.
- **Wave-energy distinction in the blast-metrics regime.** The framework adopts as a working commitment that wave and energy components of the primordial pulse propagate together for the purposes of empirical engagement. The deeper physics of the wave-energy distinction in the blast-metrics regime is an open theoretical question. The specialist input should address whether the working commitment is justifiable and what the consequences are for the framework's predictions if wave and energy components propagate at distinct rates in the blast-metrics regime.

These open questions are listed with the gating issue first and the most tractable quantitative engagement second. The remaining questions are listed in roughly descending order of physical importance to the framework. A specialist engagement that addresses the first two questions would substantially advance the framework toward making quantitative predictions.

Empirical bounds and structural consistency

The framework's commitments are constrained by existing empirical results in several places. These constraints are flagged here for the specialist's attention.

The H_0 tension and the V1.5 calculation

The framework's V1.5 attempt at engaging the H_0 tension through spatial coupling variation produced a parameter-regime bound violation against BBN and lunar-laser-ranging constraints, by roughly two orders of magnitude. This is reported as a bound violation rather than as a balanced negative result. In the parameter regime that satisfies BBN and LLR, the spatial-coupling mechanism cannot contribute to H_0 tension at observationally meaningful levels.

The framework does not currently offer an alternative mechanism for the H_0 tension. The V1.5 mechanism is ruled out; no replacement has been developed. The specialist input may identify whether the framework's commitments support any alternative mechanism for the H_0 tension, or whether the framework's position is that the H_0 tension is not addressed by UMF and remains an open question for standard cosmology.

Bounds on α variation

Atomic clock comparisons constrain the present-day rate of α variation to the 10^{-17} per year level. Quasar absorption spectroscopy (ESPRESSO sub-ppm bound, Murphy et al. 2022, and competing analyses) constrains α variation across cosmological redshifts at the 10^{-5} to 10^{-6} level over redshifts up to $z \approx 3$. BBN constraints typically give $|\Delta\alpha/\alpha| < 10^{-2}$ at $z \approx 10^9$, tighter in some analyses. The framework's prediction of variable α from variable ϵ_0 must be consistent with all three bounds. Without specification of the functional form of $\epsilon_0(T_G)$, the framework cannot calculate its predicted α variation rate, and the consistency with bounds cannot be evaluated. This is the work item the framework should attempt first (see Open theoretical questions above).

Direct detection null results

Direct dark matter detection experiments (LUX, XENON, PandaX, others) have produced increasingly tight null results across decades of searching. The framework's commitment to no-particle no-mass dark matter predicts that these experiments should never detect anything, because there are no particles to detect. The null results are consistent with the framework's commitment but do not discriminate the framework from other modified-gravity / no-particle DM programmes (MOND, TeVeS, MOG, Verlinde's emergent gravity), which make the same prediction. The framework should not cite the nulls as positive evidence; they are consistency, not support.

Radiation signatures

The framework's commitment to dark matter as non-active field effect (rather than as an active species in spacetime) predicts that there should be no radiation signatures from dark matter. The absence of detected radiation signatures (across the entire electromagnetic spectrum, in neutrinos, in gravitational waves, and in cosmic rays) is consistent with the framework's commitment. As with direct detection, this is consistency rather than discriminating support.

Inflation, the horizon problem, and the primordial power spectrum

Standard cosmology invokes inflation to address the horizon problem, the flatness problem, and the seeding of primordial structure. The framework's establishing event is gestural — the framework explicitly does not commit to specific physics in the trans-Planckian regime. This is honest but leaves the framework with no mechanism for the observed CMB homogeneity across causally disconnected patches, the observed near-flatness, or the observed scale-invariant primordial power spectrum. These are not optional features of cosmology; they are observed. The framework must either adopt inflation (and clarify how it sits inside the Region A / Region B / Region C structure) or propose an alternative mechanism for the three. "Not specified by the framework" is not a viable position for these specific observables. This is flagged for specialist input.

Framework lineage and prior empirical work

The framework lineage (UMDT and UMGH precursors) has conducted pre-registered empirical analyses that bear on UMF's empirical capacity, although the analyses were not conducted under UMF's specific commitments. The lineage analyses are referenced here as context for the framework's empirical methodology, not as direct empirical support for UMF.

Under UMDT, five pre-registered empirical analyses have been conducted: Pantheon+ residual spatial analysis (April 2026, null result at 3σ Bonferroni-corrected threshold across five tests, converted to a 1.1% upper limit on coherent fractional metric variation in the frequency band above 5.8×10^{-18} Hz); ISW cross-correlation meta-analysis (April 2026, six published measurements 2008-2022, weighted mean $A_{\text{ISW}} = 1.04 \pm 0.13$ consistent with Λ CDM at the 0.3σ level, converted to a constraint of $|\alpha| < 0.3$ at 3σ on any linear redshift trend in $A_{\text{ISW}}(z) = 1 + \alpha \cdot z$); CMB echo parametric pumping and CMB lattice Path 1 (both null); and DESI DR2 BAO analysis (April 2026, positive consistency match with all four pre-registered tests passing, with the P2 phantom-crossing preference reaching 4.2σ correlation-class significance for the DESI+CMB+DESY5 combination). The methodology across the programme is consistent: pre-registration before data inspection, Bonferroni correction for multiple testing, null reporting as primary deliverable, honest caveats.

The framework lineage's prior empirical work demonstrates capacity for pre-registered empirical analysis at the methodological standard the field expects, with the discipline of reporting nulls four times. This is context for the specialist engagement; it is not direct support for UMF's commitments. UMF's commitments differ from UMDT's (UMF commits to T_G as an independent dynamical field with the specific reflective-boundary commitment on the wavefront; UMDT v2.1 has parametrised impedance components and different interpretation A). UMF-specific empirical analyses await the resolution of the framework's open theoretical questions, with the bounded-universe topology constraint from Planck CMB low- ℓ data as the candidate first UMF-specific empirical engagement under the reflective-boundary commitment (documented in the companion document UMF Bounded-Universe Topology v1.0 at the structural-inference level).

Revision history

This document supersedes UMF v2.2 of 18 May 2026. The substantive changes are:

- **GitHub publication reference.** The framework is now publicly available at <https://github.com/OldGreyWhistle/The-UMF> as a citable reference. The current package is V2.3.
- **Entropy at the reflective wavefront added as open theoretical question.** A new open question on entropy behaviour at the reflective wavefront is added, raised by the ChatGPT comment on V2.2 (May 2026). The framework does not take a position on the answer; the question is flagged for specialist evaluation.
- **DM scaling with cosmic time question added.** A new consolidated open question on UMF's gravitoelectric DM scaling with cosmic time and its potential population-level early-universe consequences is added, coupled to the variable- α / BBN engagement and to a related new question on potential simultaneous resolution of S_8 / BBN / CMB acoustic peaks tensions through DM behaviour. Both are flagged as areas for specialist investigation rather than as framework commitments.
- **ISW meta-analysis constraint citation.** The framework lineage section now cites the specific ISW meta-analysis constraint ($|\alpha| < 0.3$ at 3σ on linear redshift trend in A_{ISW}), strengthening the framework lineage empirical context with the specific numerical constraint.
- **Bounded-universe topology positioning document referenced.** The Empirical Engagement document's candidate first UMF-specific empirical engagement (bounded-universe topology under reflective-boundary commitment) is now documented in the companion document UMF Bounded-Universe Topology v1.0. The reference is cited throughout where the bounded-universe topology question is discussed.

- **Location and timezone consistency.** Title blocks, footers, and provenance now consistently reference Central Coast NSW, AEST (Australian Eastern Standard Time, applicable in standard-time months April through October). AEDT applies in daylight-time months October through April. May 2026 is AEST.
- **Contemporary thin-shell literature reference.** The dissolving-shell candidate relaxation picture section now references contemporary peer-reviewed applications of Israel junction conditions on configurations structurally similar to UMF's setup (Schubert 2026 in Entropy, the brane cosmology literature). This strengthens the reference to Israel junction conditions as the relevant formal apparatus for the wavefront's formal description.

Closing

The framework's commitments are at the structural level rather than the formal-derivation level. The framework does not yet have quantitative predictions that distinguish it from Λ CDM at observational precision. The specialist engagement is the next step in the framework's development; the open theoretical questions listed above are the work items where the specialist's expertise is most needed.

The three cold reviews (research lead, ChatGPT, Claude Opus 4.7 incognito) converge on three substantive issues that the framework now flags explicitly: the absence of an action principle as the gating theoretical issue; the relaxation-producing- $w \approx -1$ mechanism gap as the most serious physical question; the variable-coupling Maxwell sector as the most tractable first quantitative engagement. The cold reviews also identify the same literature programmes for the framework to engage (scalar-tensor theories, Bekenstein/BSBM varying-constants formalism, Israel junction conditions for the wavefront, modified-gravity DM theories, holographic boundary formalisms). The convergence is what gives confidence that these are genuine framework issues rather than artefacts of any single reviewer.

The framework should be evaluated for whether its structural commitments are physically defensible, whether they are internally consistent, and whether the open theoretical questions are tractable. A specialist who finds the structural commitments unsound can stop there; there is no point evaluating the formal scaffold if the foundations are rejected. A specialist who finds the structural commitments at least plausible should consider which of the open theoretical questions is most tractable to address, and what additional commitments would need to be made before quantitative predictions can be developed. Both cold reviews recommend the variable-coupling Maxwell sector as the first such engagement.